

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

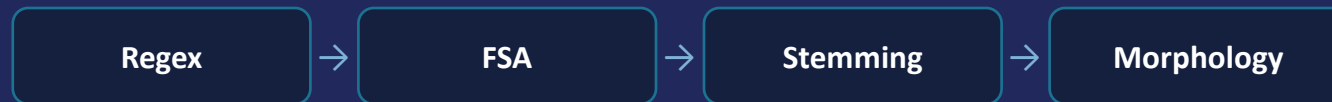
Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Integrated NLP Pipeline

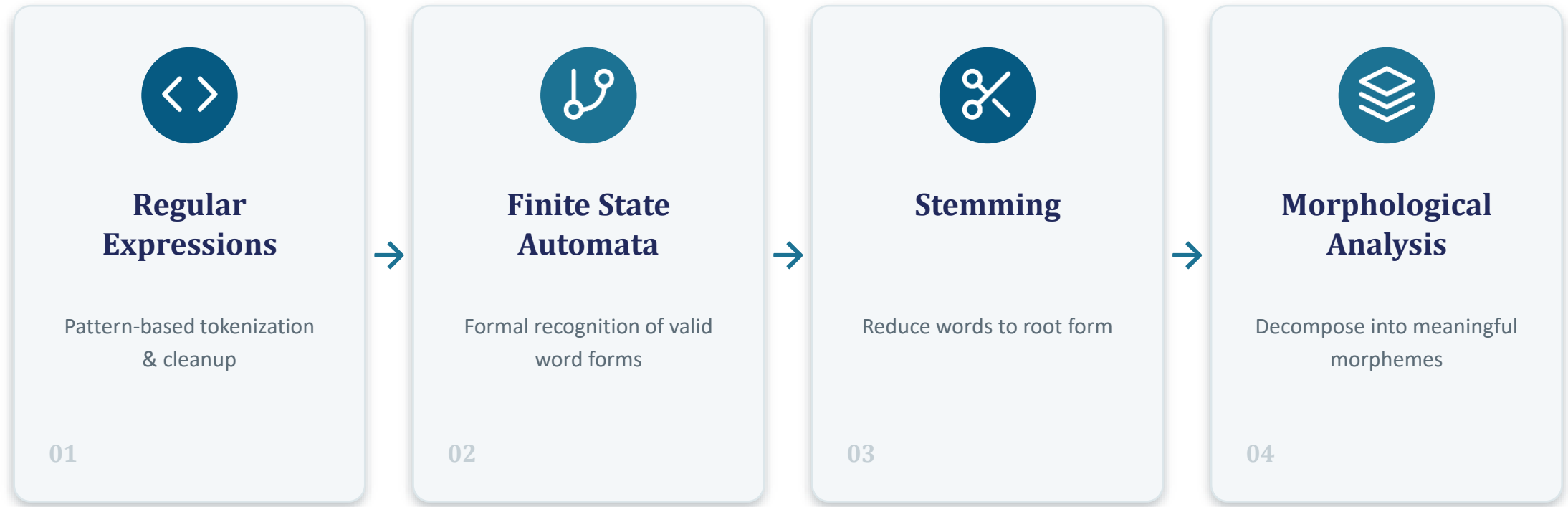
From Regular Expressions to Morphological Processing

Demonstrating a complete text-preprocessing pipeline that unifies regular expressions, finite state automata, stemming, and morphological analysis.



Pipeline Overview

Raw text moves through four cooperating stages before reaching a structured, analyzable form.





STAGE 01

Regular Expressions

The first filter: pattern-matching rules that clean, split, and normalize raw text before any linguistic reasoning begins.

Core Responsibilities



Tokenization

Split raw input into candidate words using patterns like `\w+` or `\S+`.



Noise removal

Strip punctuation, HTML tags, and extra whitespace with substitution rules.



Normalization

Lowercase text, collapse contractions, and standardize numerals or dates.



Boundary detection

Identify sentence and clause boundaries using lookahead/lookbehind patterns.

EXAMPLE PATTERN

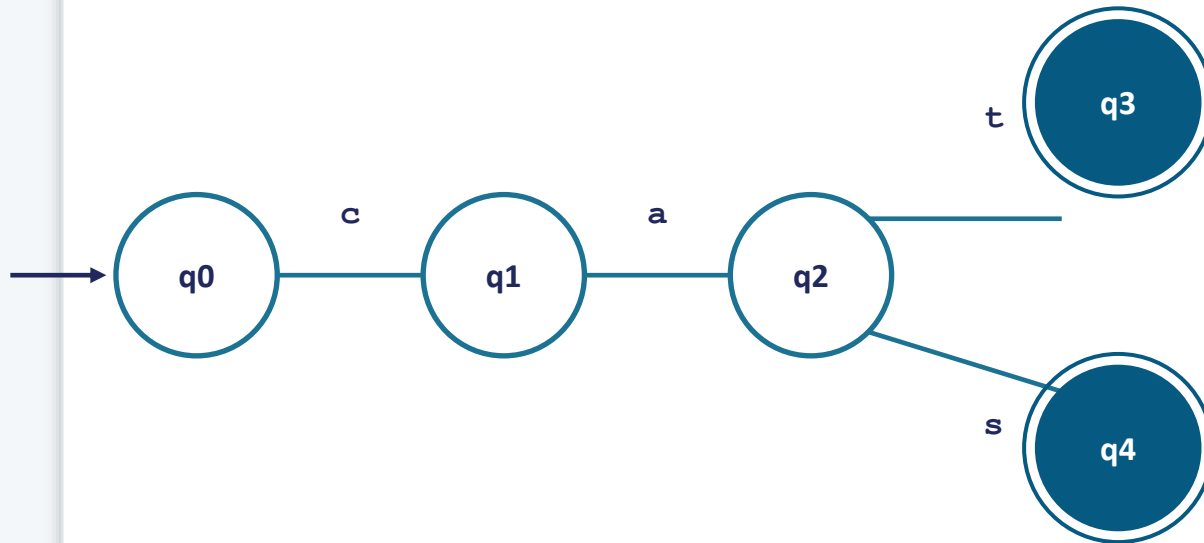
```
re.findall(r'\b[A-Za-z]+(?:\'[a-z]+)?\b', text)
```

Matches words, keeping internal apostrophes (e.g. "don't") intact as single tokens.

Finite State Automata

A formal machine of states and transitions that recognizes whether a token is a well-formed word — the theoretical backbone beneath regex engines and morphological rules.

AUTOMATON FOR RECOGNIZING "cats" / "cat"



q3 accepts "cat" · q4 accepts "cats" · double ring = accepting state

Deterministic

Each state has exactly one transition per input symbol — fast, predictable matching.

Composable

Small automata for prefixes, roots, and suffixes combine into a full morphological analyzer.

Regex $\equiv$ FSA

Every regular expression compiles to an equivalent finite state machine internally.

Stemming

A fast, rule-based reduction of inflected words to an approximate root — trading linguistic precision for speed.

Porter Stemmer — Rule Cascade

1

Remove plurals & -ed / -ing suffixes (SSES→SS, IES→I, ING→ε)

2

Simplify double suffixes (ATIONAL→ATE, IZATION→IZE)

3

Reduce remaining suffixes (ICATE→IC, FUL→ε)

4

Strip residual suffixes on longer stems (-ANCE, -ENT, -ION)

BEFORE → AFTER

relational	relate
conditional	condition
argument	argue
generalization	general
happiness	happi
studies	studi

Morphological Analysis

Where stemming trims heuristically, morphological analysis decomposes a word into its complete, linguistically valid set of morphemes.

un

prefix

happi

root

ness

suffix

es

inflection

"unhappiness" — analyzed as prefix + bound root + derivational suffix

TWO-LEVEL MORPHOLOGY

Lexical level: fox + N + PL

↕ transducer rules

Surface level: f o x e s

Finite state transducers (FSTs) map lexical morpheme sequences to their surface spelling and back — the same automaton theory from Stage 2, now bidirectional.

Inflectional

Same lexeme, grammatical variation — tense, number, case (walk → walked).

Derivational

New lexeme, new meaning or part of speech (happy → happiness).

Integrated Walkthrough

Tracing a single sentence through all four pipeline stages, end to end.

INPUT "The runners quickly organized unbelievable races."



Regex

[The, runners, quickly, organized, unbelievable, races]

Tokenized on \w+, punctuation stripped



FSA

All 6 tokens accepted as valid English word-forms

Each token traced through the word-recognition automaton



Stemming

[run, runner, quick, organ, unbeliev, race]

Porter rules strip -ers, -ly, -ized, -able (heuristic, approximate)



Morphology

organized → organize+PAST · unbelievable → un+believe+able

Full morpheme decomposition with grammatical roles

Why Integrate the Stages?

Each technique alone is limited; combined, they trade off speed, precision, and linguistic depth.



Regex alone

STRENGTH

Extremely fast, simple to write

LIMITATION

No understanding of word structure or meaning



Stemming alone

STRENGTH

Lightweight, good for search & indexing

LIMITATION

Produces non-words; ignores grammar (e.g. "studi")



Morphology alone

STRENGTH

Linguistically accurate, reversible

LIMITATION

Computationally heavier, needs a full lexicon



Chaining regex → FSA → stemming → morphology lets each stage compensate for the previous one's blind spots.

Applications

The same pipeline pattern underlies a wide range of downstream NLP systems.



Search Engines

Stemming expands query recall by matching word-form variants.



Spell Checkers

FSA-based dictionaries validate word forms in real time.



Machine Translation

Morphological analysis aligns inflected forms across languages.



Text Classification

Regex + stemming normalize features before vectorization.



Speech Recognition

Morpheme lexicons constrain the space of valid output words.



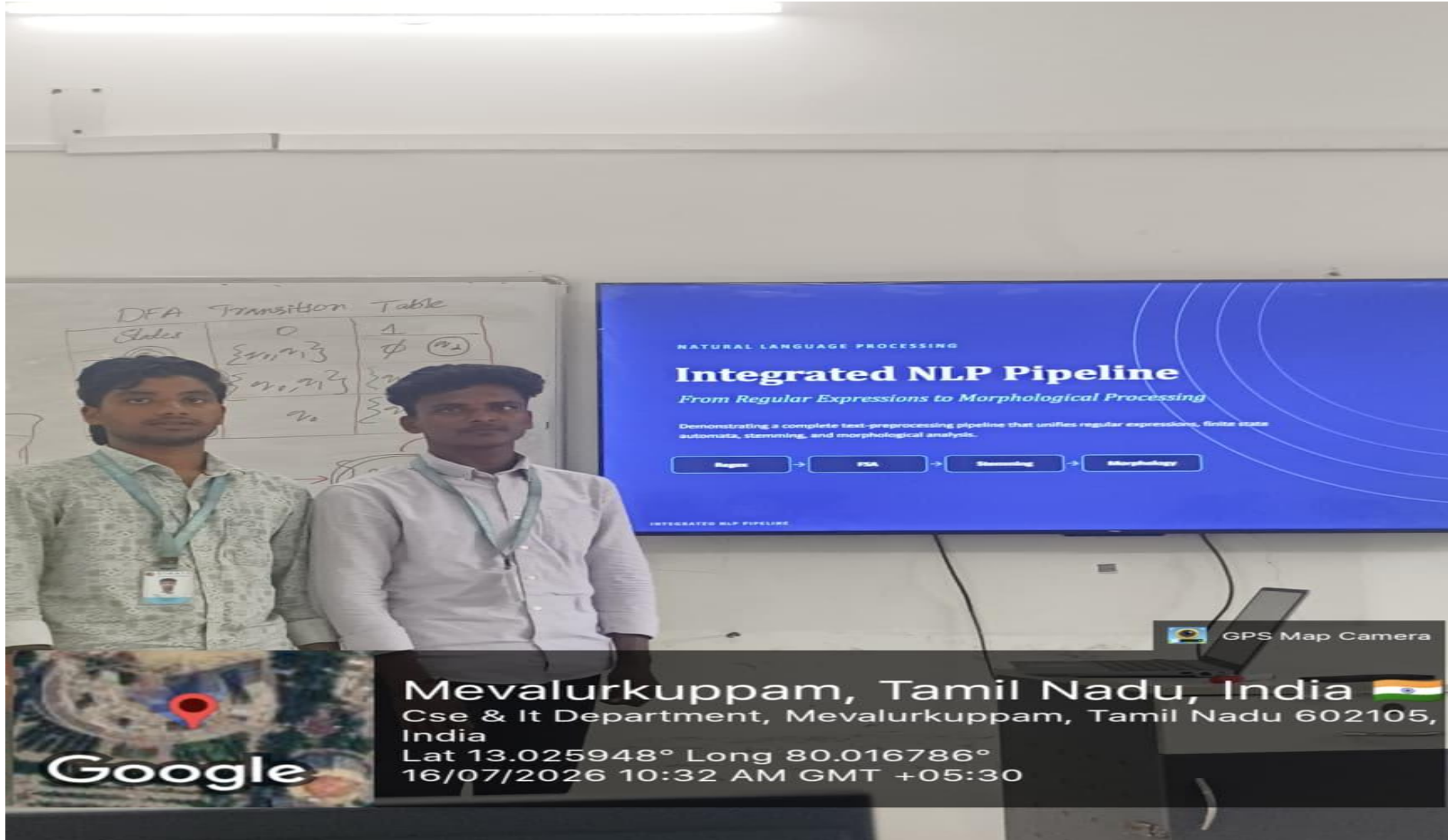
Information Extraction

Pattern rules isolate entities before deeper linguistic parsing.

Key Takeaways

- Regex handles surface-level pattern matching — fast but linguistically blind.
- Finite state automata formalize word recognition and underlie both regex and morphology engines.
- Stemming offers a lightweight approximation of a word's root, at the cost of precision.
- Morphological analysis provides complete, structured decomposition into real morphemes.
- Integrated, the four stages form a pipeline that is fast, formally grounded, and linguistically accurate.

Thank you



DFA Transition Table

States	0	1
$\{q_0, q_1\}$	$\{q_1, q_2\}$	ϕ (q_2)
q_0	q_1	q_2

NATURAL LANGUAGE PROCESSING

Integrated NLP Pipeline

From Regular Expressions to Morphological Processing

Demonstrating a complete text-preprocessing pipeline that unifies regular expressions, finite state automata, stemming, and morphological analysis.



INTEGRATED NLP PIPELINE

GPS Map Camera

Google

Mevalurkuppam, Tamil Nadu, India 🇮🇳

Cse & It Department, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.025948° Long 80.016786°

16/07/2026 10:32 AM GMT +05:30